```
import pandas as pd

# Load dataset splits
train = pd.read_csv("train.txt", sep="\t", names=["head","relation","tail"])
valid = pd.read_csv("valid.txt", sep="\t", names=["head","relation","tail"])
test = pd.read_csv("test.txt", sep="\t", names=["head","relation","tail"])

print("Train shape:", train.shape)
print("Valid shape:", valid.shape)
print("Test shape:", test.shape)
```

```
Train shape: (120656, 3)
Valid shape: (17535, 3)
Test shape: (20466, 3)
```

```
# Remove nulls
train = train.dropna().drop_duplicates().reset_index(drop=True)
valid = valid.dropna().drop_duplicates().reset_index(drop=True)
test = test.dropna().drop_duplicates().reset_index(drop=True)
```

```
def feature_engineering(df):

    # Length features
    df["head_length"] = df["head"].apply(len)
    df["tail_length"] = df["tail"].apply(len)
    df["relation_length"] = df["relation"].apply(len)

    # Word count
    df["head_word_count"] = df["head"].apply(lambda x: len(x.split("_")))
    df["tail_word_count"] = df["tail"].apply(lambda x: len(x.split("_")))
    df["relation_word_count"] = df["relation"].apply(lambda x: len(x.split("_")))

    # Uppercase count
    df["head_upper"] = df["head"].apply(lambda x: sum(c.isupper() for c in x))
    df["tail_upper"] = df["tail"].apply(lambda x: sum(c.isupper() for c in x))
    df["relation_upper"] = df["relation"].apply(lambda x: sum(c.isupper() for c in x))

    # Digit count
    df["head_digit_count"] = df["head"].apply(lambda x: sum(c.isdigit() for c in x))
    df["tail_digit_count"] = df["tail"].apply(lambda x: sum(c.isdigit() for c in x))
    df["relation_digit_count"] = df["relation"].apply(lambda x: sum(c.isdigit() for c in x))

    # Has number
    df["head_has_number"] = df["head_digit_count"].apply(lambda x: 1 if x > 0 else 0)
    df["tail_has_number"] = df["tail_digit_count"].apply(lambda x: 1 if x > 0 else 0)
    df["relation_has_number"] = df["relation_digit_count"].apply(lambda x: 1 if x > 0 else 0)

    return df

train = feature_engineering(train)
valid = feature_engineering(valid)
test = feature_engineering(test)
```

```
# Compute degrees using training data only
head_degree = train["head"].value_counts()
tail_degree = train["tail"].value_counts()
relation_freq = train["relation"].value_counts()

def add_degree_features(df):
    df["head_degree"] = df["head"].map(head_degree).fillna(0)
    df["tail_degree"] = df["tail"].map(tail_degree).fillna(0)
    df["relation_frequency"] = df["relation"].map(relation_freq).fillna(0)
    df["degree_difference"] = df["head_degree"] - df["tail_degree"]
    df["head_tail_length_diff"] = df["head_length"] - df["tail_length"]
    df["head_tail_length_sum"] = df["head_length"] + df["tail_length"]
    return df

train = add_degree_features(train)
valid = add_degree_features(valid)
test = add_degree_features(test)
```

```
from sklearn.preprocessing import LabelEncoder

# Combine all data for fitting encoders
all_heads = pd.concat([train["head"], valid["head"], test["head"]])
all_tails = pd.concat([train["tail"], valid["tail"], test["tail"]])
all_relations = pd.concat([train["relation"], valid["relation"], test["relation"]])
```

```
# Create encoders
le_head = LabelEncoder()
le_tail = LabelEncoder()
le_relation = LabelEncoder()

# Fit on all data
le_head.fit(all_heads)
le_tail.fit(all_tails)
le_relation.fit(all_relations)

# Transform each dataset
for df in [train, valid, test]:
    df["head_enc"] = le_head.transform(df["head"])
    df["tail_enc"] = le_tail.transform(df["tail"])
    df["relation_enc"] = le_relation.transform(df["relation"])
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

# Select numeric columns except target
feature_cols = train.select_dtypes(include=["int64","float64"]).columns
feature_cols = feature_cols.drop("relation_enc")

# Fit on train only
train[feature_cols] = scaler.fit_transform(train[feature_cols])

# Transform valid & test
valid[feature_cols] = scaler.transform(valid[feature_cols])
test[feature_cols] = scaler.transform(test[feature_cols])
```

```
X_train = train.drop(columns=["head","tail","relation","relation_enc"])
y_train = train["relation_enc"]

X_valid = valid.drop(columns=["head","tail","relation","relation_enc"])
y_valid = valid["relation_enc"]

X_test = test.drop(columns=["head","tail","relation","relation_enc"])
y_test = test["relation_enc"]

print("Training shape:", X_train.shape)
print("Testing shape:", X_test.shape)
```

```
Training shape: (120655, 23)
Testing shape: (20466, 23)
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model_lr = LogisticRegression(max_iter=1000)
model_lr.fit(X_train, y_train)

pred_lr = model_lr.predict(X_test)

print("Logistic Regression Accuracy:",
      accuracy_score(y_test, pred_lr))
```

```
Logistic Regression Accuracy: 0.7560343985146096
```

```
from sklearn.ensemble import RandomForestClassifier

model_rf = RandomForestClassifier(n_estimators=100)
model_rf.fit(X_train, y_train)

pred_rf = model_rf.predict(X_test)

print("Random Forest Accuracy:",
      accuracy_score(y_test, pred_rf))
```

```
Random Forest Accuracy: 0.9922798788234144
```

```
from sklearn.metrics import classification_report
import warnings
warnings.filterwarnings("ignore")

print(classification_report(y_test, pred_rf, zero_division=0))
```

```
           185       0.78      0.75      0.77        24
           186       1.00      1.00      1.00       436
           187       0.99      0.98      0.98        98
           188       1.00      1.00      1.00       494
           189       0.99      1.00      0.99       170
           190       1.00      1.00      1.00       305
           191       1.00      1.00      1.00      1311
           192       0.99      1.00      1.00       127
           193       0.88      1.00      0.93         7
           194       0.94      0.97      0.96        35
           195       0.80      1.00      0.89         4
           196       1.00      1.00      1.00       346
           197       0.83      0.77      0.80        13
           198       1.00      1.00      1.00         9
           199       1.00      1.00      1.00        10
           200       1.00      1.00      1.00        12
           201       1.00      1.00      1.00        16
           202       1.00      1.00      1.00        43
           203       0.99      0.98      0.98        86
           204       1.00      0.94      0.97        17
           205       1.00      1.00      1.00        16
           206       1.00      1.00      1.00        42
           209       1.00      1.00      1.00        22
           210       1.00      0.99      0.99        77
           211       1.00      1.00      1.00         1
           212       1.00      0.67      0.80         3
           213       1.00      1.00      1.00         3
           214       1.00      1.00      1.00         2
           215       0.98      0.98      0.98        53
           216       0.90      0.89      0.90        53
           217       1.00      0.91      0.95        11
           218       0.91      0.94      0.93        34
           219       1.00      1.00      1.00        60
           220       0.95      1.00      0.98        20
           221       1.00      1.00      1.00         7
           222       1.00      0.92      0.96        26
           223       1.00      0.82      0.90        11
           224       1.00      1.00      1.00        22
           225       1.00      1.00      1.00         4
           226       0.97      1.00      0.98        32
           227       0.98      0.99      0.98        94
           228       1.00      1.00      1.00        24
           229       1.00      0.50      0.67         2
           230       0.99      1.00      0.99        91
           231       1.00      1.00      1.00        11
           232       1.00      1.00      1.00         9
           233       1.00      0.89      0.94         9
           234       0.92      1.00      0.96        11
           235       0.96      1.00      0.98        24
           236       1.00      1.00      1.00        50

    accuracy                           0.99     20466
   macro avg       0.96      0.94      0.95     20466
weighted avg       0.99      0.99      0.99     20466
```

```
!pip install neo4j tqdm
```

```
Collecting neo4j
  Downloading neo4j-6.1.0-py3-none-any.whl.metadata (5.3 kB)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (4.67.3)
Requirement already satisfied: pytz in /usr/local/lib/python3.12/dist-packages (from neo4j) (2025.2)
Downloading neo4j-6.1.0-py3-none-any.whl (325 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 325.3/325.3 kB 6.6 MB/s eta 0:00:00
Installing collected packages: neo4j
Successfully installed neo4j-6.1.0
```

```
from neo4j import GraphDatabase

uri = "neo4j+s://71995882.databases.neo4j.io"
username = "71995882"
password = "UuSgt9NWfJY0_4NCzsQlsbSjnr662VgEkxEly5gV2vY"

driver = GraphDatabase.driver(uri, auth=(username, password))

print("Connected successfully!")
```

```
Connected successfully!
```

```
df.head()
print(len(df))
```

```
20466
```

```
from tqdm import tqdm

def insert_batch(tx, batch):
```

```
    query = """
    UNWIND $rows AS row
    MERGE (h:Entity {name: row.head})
    MERGE (t:Entity {name: row.tail})
    MERGE (h)-[:RELATION {type: row.relation}]->(t)
    """
    tx.run(query, rows=batch)

batch_size = 1000
records = df.to_dict("records")

with driver.session() as session:
    for i in tqdm(range(0, len(records), batch_size)):
        batch = records[i:i+batch_size]
        session.execute_write(insert_batch, batch)

print("Data inserted successfully!")
```

```
100%|██████████| 21/21 [03:16<00:00,  9.37s/it]Data inserted successfully!
```

```
!pip install spacy neo4j
!python -m spacy download en_core_web_trf
```

```
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: pytz in /usr/local/lib/python3.12/dist-packages (from neo4j) (2025.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->sp
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-sl
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Collecting en-core-web-trf==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_trf-3.8.0/en_core_web_trf-3.8.0-py3-
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 457.4/457.4 MB 808.9 kB/s eta 0:00:00
Collecting spacy-curated-transformers<1.0.0,>=0.2.2 (from en-core-web-trf==3.8.0)
  Downloading spacy_curated_transformers-0.3.1-py2.py3-none-any.whl.metadata (2.7 kB)
Collecting curated-transformers<0.2.0,>=0.1.0 (from spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0)
  Downloading curated_transformers-0.1.1-py2.py3-none-any.whl.metadata (965 bytes)
Collecting curated-tokenizers<0.1.0,>=0.0.9 (from spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0)
  Downloading curated_tokenizers-0.0.9-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.9 kB)
Requirement already satisfied: torch>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from spacy-curated-transformers<1
Requirement already satisfied: regex>=2022 in /usr/local/lib/python3.12/dist-packages (from curated-tokenizers<0.1.0,>=0.0
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-tran
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->s
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-tr
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curat
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transf
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.12.0->spa
Downloading spacy_curated_transformers-0.3.1-py2.py3-none-any.whl (237 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 237.9/237.9 kB 4.9 MB/s eta 0:00:00
Downloading curated_tokenizers-0.0.9-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (734 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 734.0/734.0 kB 23.3 MB/s eta 0:00:00
Downloading curated_transformers-0.1.1-py2.py3-none-any.whl (25 kB)
Installing collected packages: curated-tokenizers, curated-transformers, spacy-curated-transformers, en-core-web-trf
Successfully installed curated-tokenizers-0.0.9 curated-transformers-0.1.1 en-core-web-trf-3.8.0 spacy-curated-transformer
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_trf')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy

# Load transformer-based NER model
nlp = spacy.load("en_core_web_trf")
```

```
print("Model Loaded Successfully")
```

```
Model Loaded Successfully
```

```
text = """
Barack Obama was born in Hawaii.
He was the President of the United States.
Microsoft was founded by Bill Gates in Albuquerque.
Google is headquartered in California.
"""
```

```
doc = nlp(text)

entities = []

for ent in doc.ents:
    print(ent.text, "->", ent.label_)
    entities.append((ent.text, ent.label_))
```

```
Barack Obama -> PERSON
Hawaii -> GPE
the United States -> GPE
Microsoft -> ORG
Bill Gates -> PERSON
Albuquerque -> GPE
Google -> ORG
California -> GPE
```

```
entities = list(set(entities))
print(entities)
```

```
[('Barack Obama', 'PERSON'), ('Bill Gates', 'PERSON'), ('the United States', 'GPE'), ('Albuquerque', 'GPE'), ('California',
```

```
from neo4j import GraphDatabase

uri = "neo4j+s://71995882.databases.neo4j.io"
username = "71995882"
password = "UuSgt9NWfJY0_4NCzsQlsbSjnr662VgEkxEly5gV2vY"

driver = GraphDatabase.driver(uri, auth=(username, password))
```

```
def create_entity(tx, name, label):
    query = f"MERGE (e:{label} {{name: $name}})"
    tx.run(query, name=name)

with driver.session() as session:
    for name, label in entities:
        session.execute_write(create_entity, name, label)

print("Entities Stored Successfully")
```

```
Entities Stored Successfully
```

```
def create_relationship(tx, e1, e2):
    query = """
    MATCH (a {name: $e1})
    MATCH (b {name: $e2})
    MERGE (a)-[:RELATED_TO]->(b)
    """
    tx.run(query, e1=e1, e2=e2)

sentences = list(doc.sents)

with driver.session() as session:
    for sent in sentences:
        sent_entities = [ent.text for ent in sent.ents]
        for i in range(len(sent_entities)):
            for j in range(i+1, len(sent_entities)):
                session.execute_write(
                    create_relationship,
                    sent_entities[i],
                    sent_entities[j]
                )

print("Relationships Created Successfully")
```

```
Relationships Created Successfully
```

```
!pip install sentence-transformers pinecone
```

```
Collecting pinecone
  Downloading pinecone-8.1.0-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-trans
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transform
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transfor
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.67.3)
Requirement already satisfied: certifi>=2019.11.17 in /usr/local/lib/python3.12/dist-packages (from pinecone) (2026.2.25)
Requirement already satisfied: orjson>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from pinecone) (3.11.7)
Collecting pinecone-plugin-assistant<4.0.0,>=3.0.1 (from pinecone)
  Downloading pinecone_plugin_assistant-3.0.3-py3-none-any.whl.metadata (30 kB)
Collecting pinecone-plugin-interface<0.1.0,>=0.0.7 (from pinecone)
  Downloading pinecone_plugin_interface-0.0.7-py3-none-any.whl.metadata (1.2 kB)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from pinecone) (2.9.0.po
Requirement already satisfied: urllib3>=1.26.5 in /usr/local/lib/python3.12/dist-packages (from pinecone) (2.5.0)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->s
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sente
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-tr
Requirement already satisfied: requests<3.0.0,>=2.32.3 in /usr/local/lib/python3.12/dist-packages (from pinecone-plugin-as
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.5.3->pinecone)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transfo
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<6
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->se
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.
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Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentenc
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sen
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.20.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.20.
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Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Downloading pinecone-8.1.0-py3-none-any.whl (742 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 742.7/742.7 kB 13.5 MB/s eta 0:00:00
Downloading pinecone_plugin_assistant-3.0.3-py3-none-any.whl (280 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 280.7/280.7 kB 20.6 MB/s eta 0:00:00
Downloading pinecone_plugin_interface-0.0.7-py3-none-any.whl (6.2 kB)
Installing collected packages: pinecone-plugin-interface, pinecone-plugin-assistant, pinecone
Successfully installed pinecone-8.1.0 pinecone-plugin-assistant-3.0.3 pinecone-plugin-interface-0.0.7
```

```
from sentence_transformers import SentenceTransformer
import pinecone
import numpy as np
```

```
from pinecone import Pinecone, ServerlessSpec
```

```
pc = Pinecone(api_key="pcsk_2HwvL3_PuG7ed7Zp3c6hTykLV5ojHxz8wa7ikPusqbSbm8Vz6bjXyGnsqEcgZKuofyVZ4N")
```

```
from pinecone import Pinecone, ServerlessSpec

# Initialize your client
pc = Pinecone(api_key="pcsk_2HwvL3_PuG7ed7Zp3c6hTykLV5ojHxz8wa7ikPusqbSbm8Vz6bjXyGnsqEcgZKuofyVZ4N")

index_name = "semantic-search-index"

if index_name not in pc.list_indexes().names():
    pc.create_index(
        index=index_name,
        dimension=384,
        metric='cosine',
        spec=ServerlessSpec(
            cloud='aws',
            region='us-east-1'
```

```
        )
    )
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Loading weights: 100%                                                    103/103 [00:00<00:00, 391.28it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
documents = [
    "Barack Obama was the President of the United States",
    "Microsoft was founded by Bill Gates",
    "Google is headquartered in California",
    "Tesla was founded by Elon Musk",
    "Amazon is an e-commerce company"
]
```

```
embeddings = model.encode(documents)
```

```
from pinecone import Pinecone

# 1. Initialize Pinecone with your API Key
pc = Pinecone(api_key="pcsk_2HwvL3_PuG7ed7Zp3c6hTykLV5ojHxz8wa7ikPusqbSbm8Vz6bjXyGnsqEcgZKuofyVZ4N")

# 2. Define the 'index' variable by connecting to your specific index name
# This is the step that is likely missing!
index = pc.Index("semantic-search-index")

# Now your existing code will work:
# index.upsert(vectors)
```

```
vectors = []

for i, emb in enumerate(embeddings):
    vectors.append(
        (str(i), emb.tolist())
    )

index.upsert(vectors)

print("Embeddings Stored in Pinecone")
```

```
Embeddings Stored in Pinecone
```

```
query = "Who founded Microsoft?"

query_embedding = model.encode([query])

results = index.query(
    vector=query_embedding[0].tolist(),
    top_k=3
)

print(results)
```

```
QueryResponse(matches=[{'id': '1', 'score': 0.903079033, 'values': []}, {'id': '3', 'score': 0.370456725, 'values': []}, {'i
```

```
for match in results['matches']:
    idx = int(match['id'])
    print(documents[idx])
```

```
Microsoft was founded by Bill Gates
Tesla was founded by Elon Musk
Google is headquartered in California
```